

Visualising spatial sound in augmented reality: visualisation techniques for locating sound



Joe Weir



Problem

- Situations arise where peoples hearing becomes impaired
 - Pedestrians wearing noise cancelling headphones
 - Workers wearing hearing protection in loud factories
- Impacts situational awareness
 - Isolates people from their surroundings
 - Safety concerns

Background

- Wayfinding in augmented reality
- Out of view object visualisation
- Gap
 - No current design guidelines for visualising spatial sound in augmented reality
 - Visualisations in wayfinding and out of view object visualisation do not show the localisation techniques uncertainty

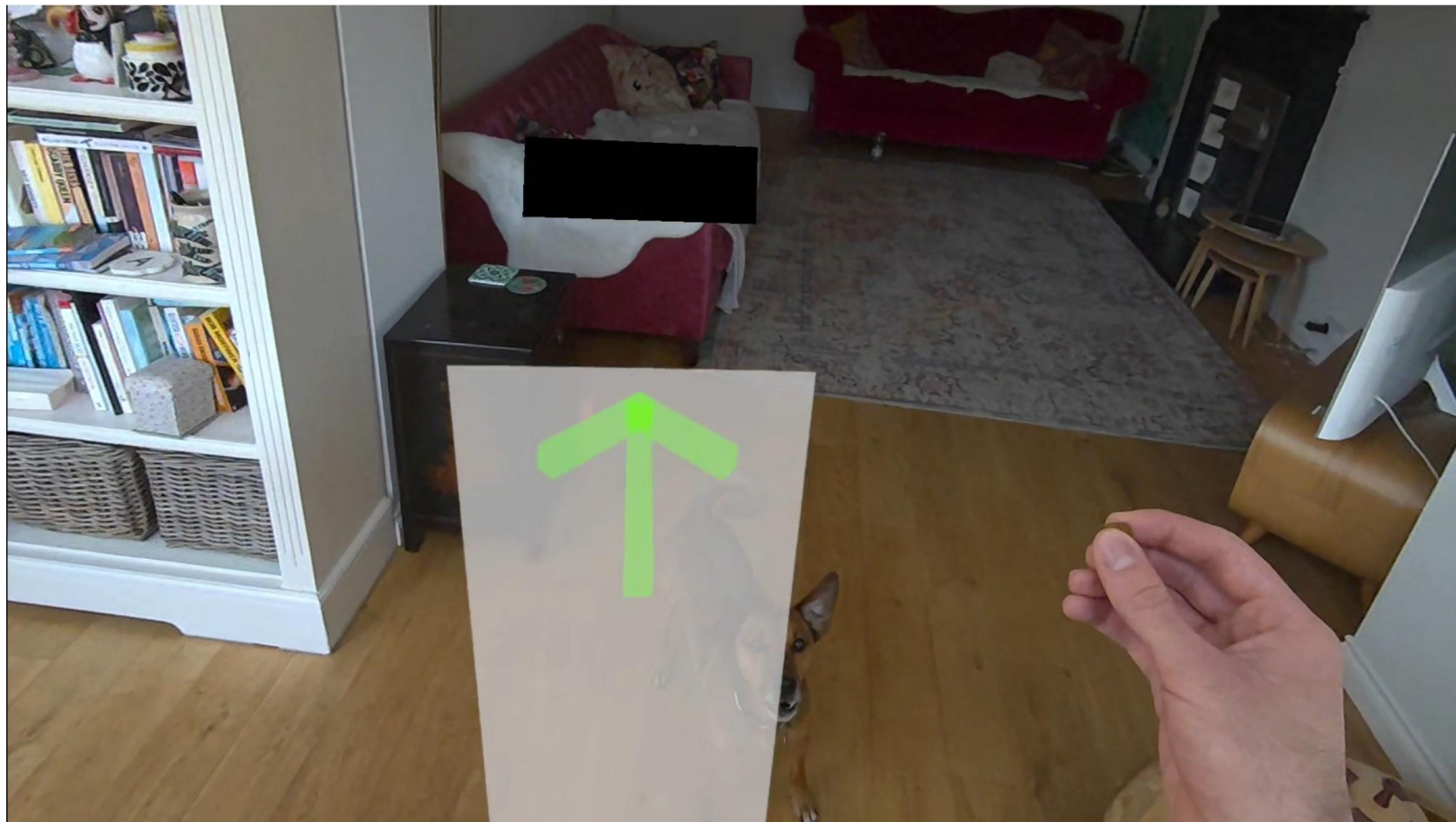
Solution

- Using a microphone array to estimate the direction and proxy distance of sound
- Visualise the user's aural environment using augmented reality
- Design and evaluate multiple visualisation techniques that encode the microphone arrays sensing uncertainty
- Evaluate which visualisations are effective at conveying one's aural environment

Visualisations Created

- Visualisations derived from visualisation techniques from wayfinding and out of view object visualisation literature
- Arrow
- Radar
- Lights

Arrow



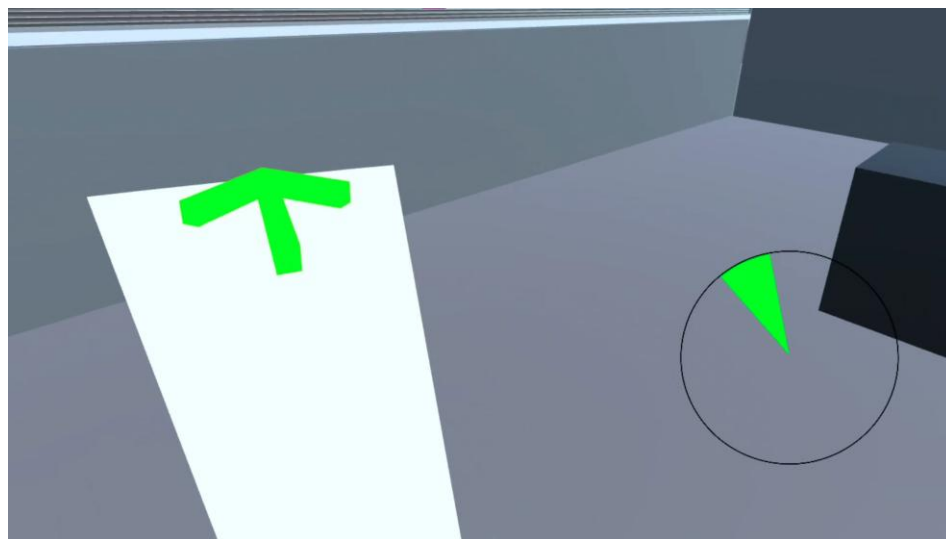
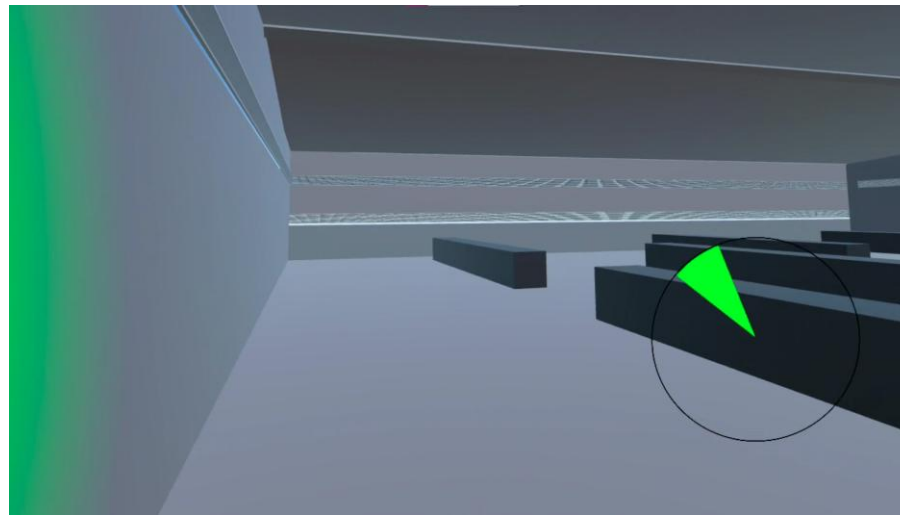
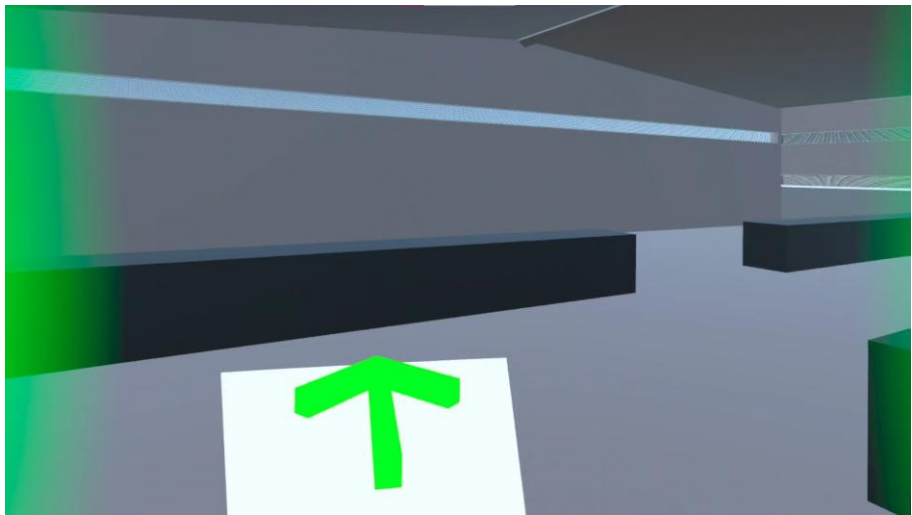
Radar



Lights



Combinations

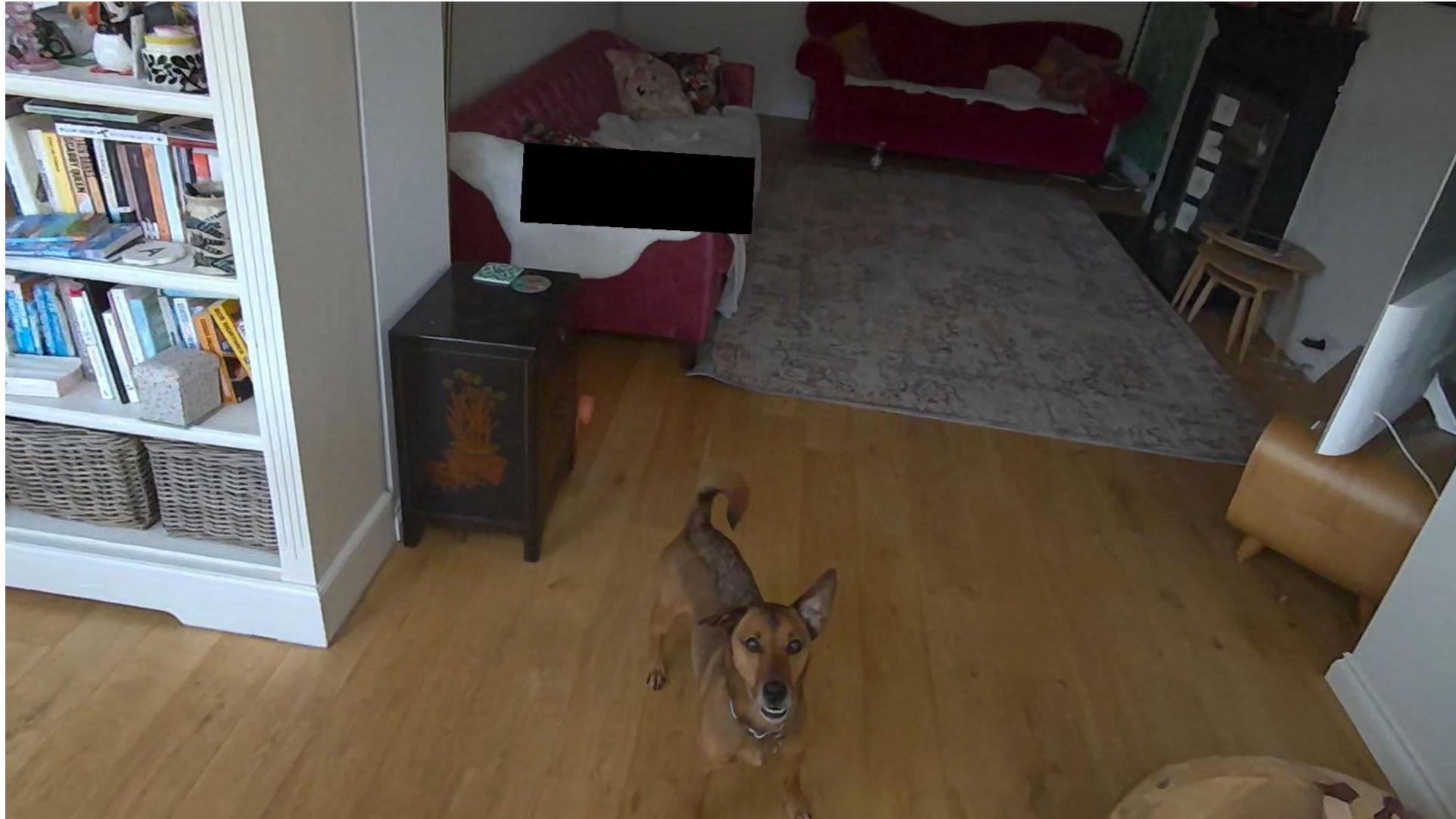


Prototype

- Using microphone array mounted on top of mixed reality headset



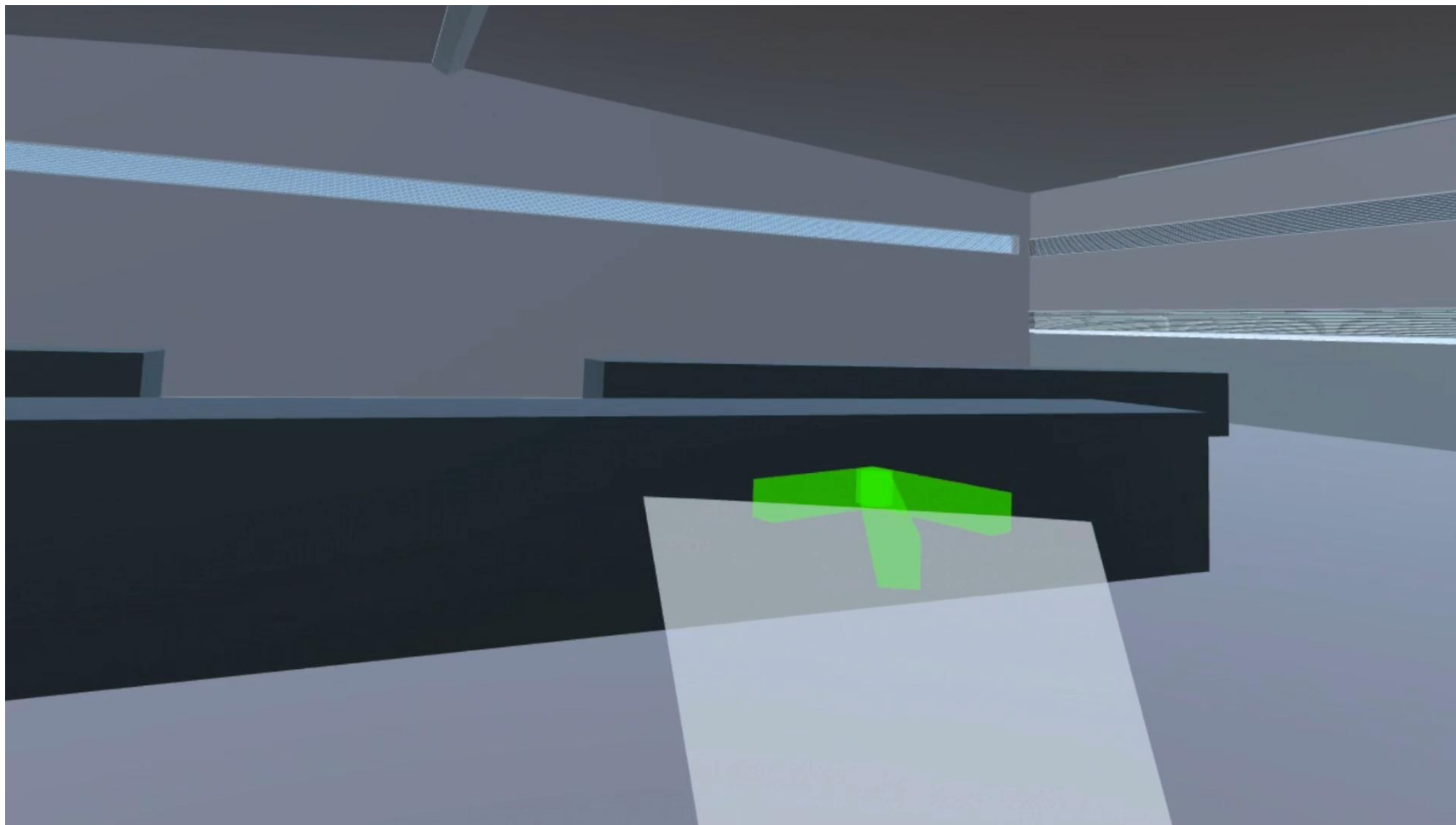
Prototype



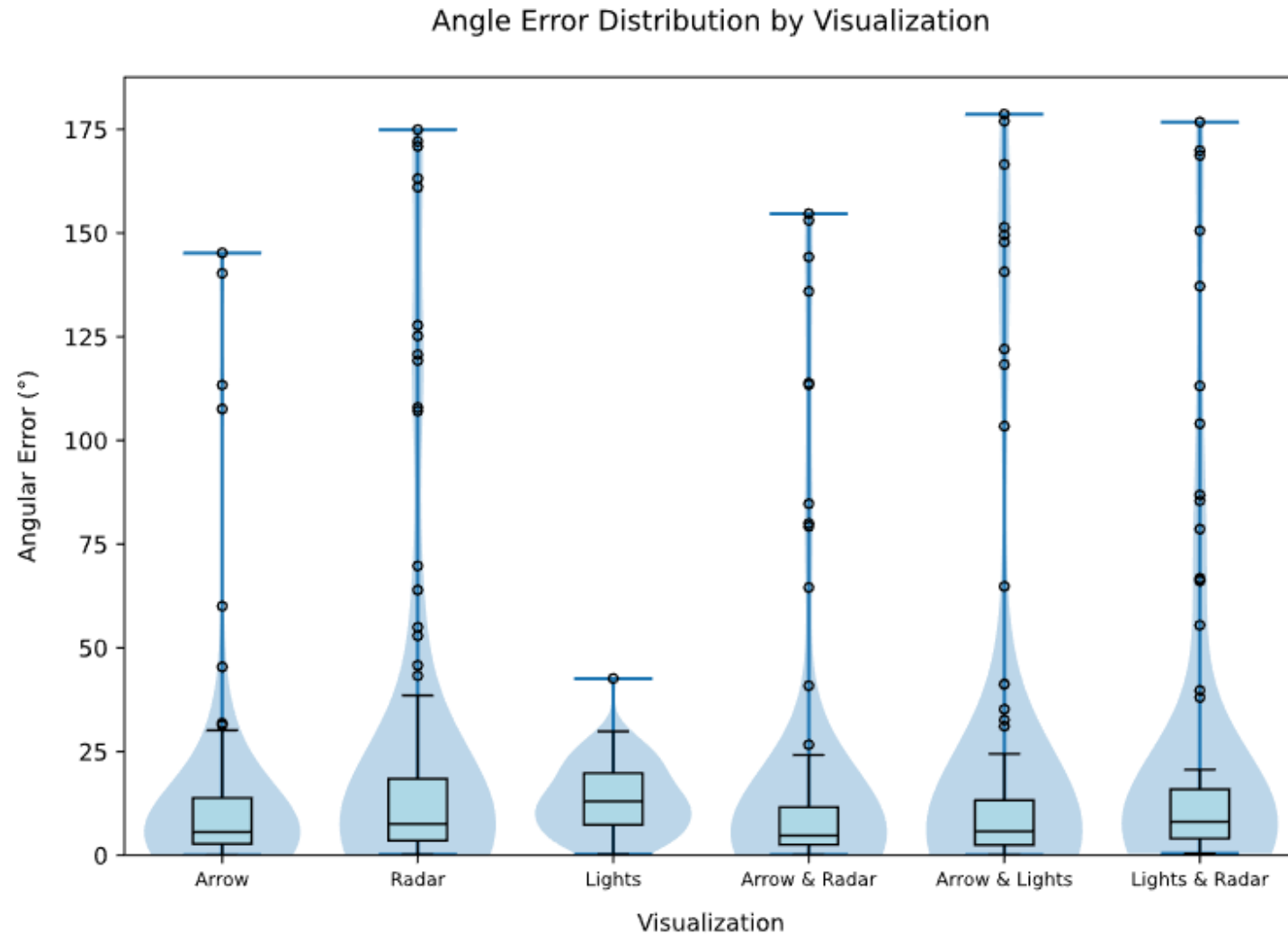
Experiment

- Users located sound sources in a virtual environment using each of the visualisations
- Uncertainty of the microphone array was simulated
- Collected quantitative and qualitative data through performance metrics, surveys and semi-structured interviews
- Measured:
 - Accuracy (distance + angle)
 - Perceived workload (NASA TLX) and Perceived Visual Demand/Load
 - User preference and use cases and improvements of the system

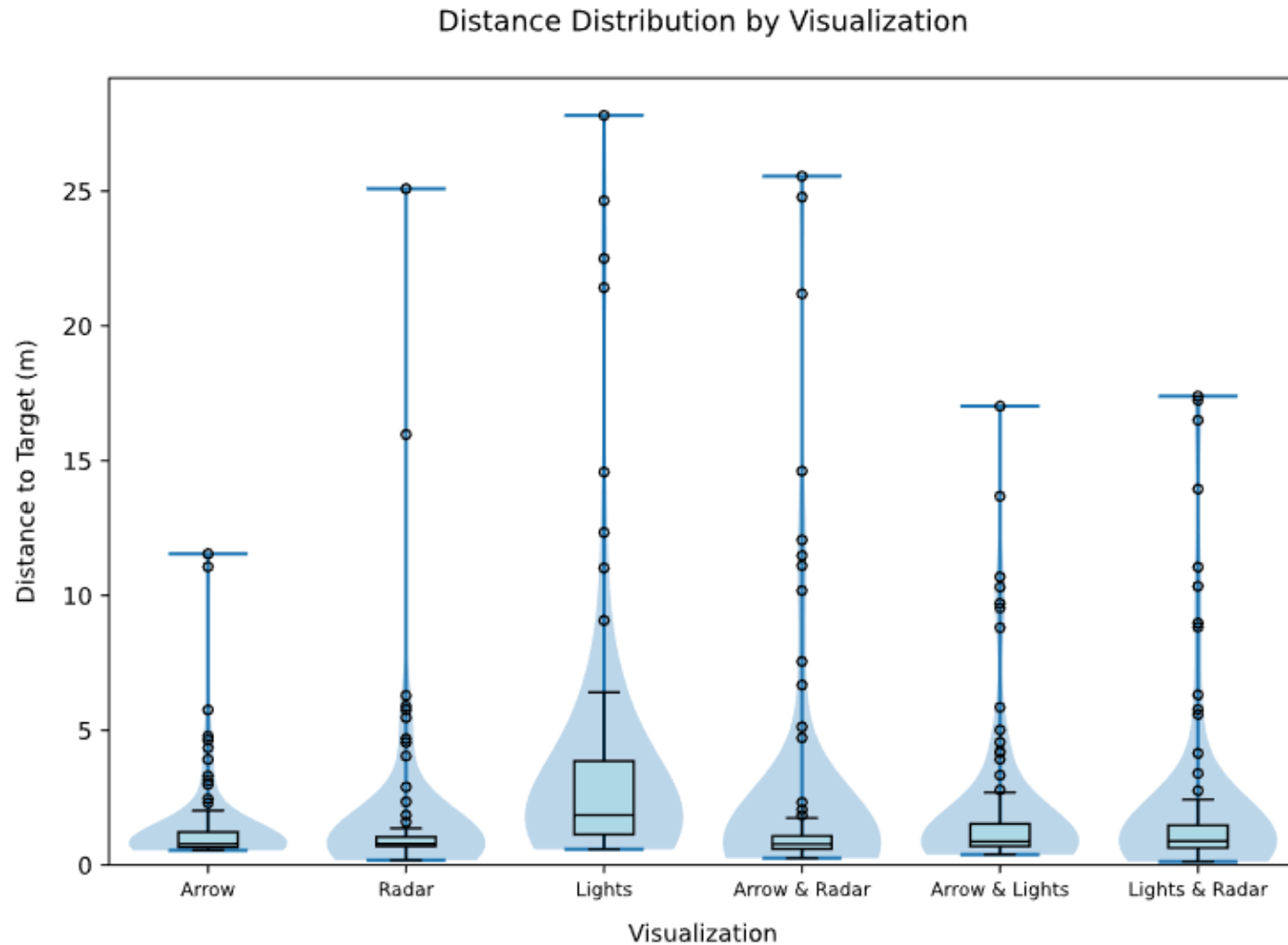
Experiment



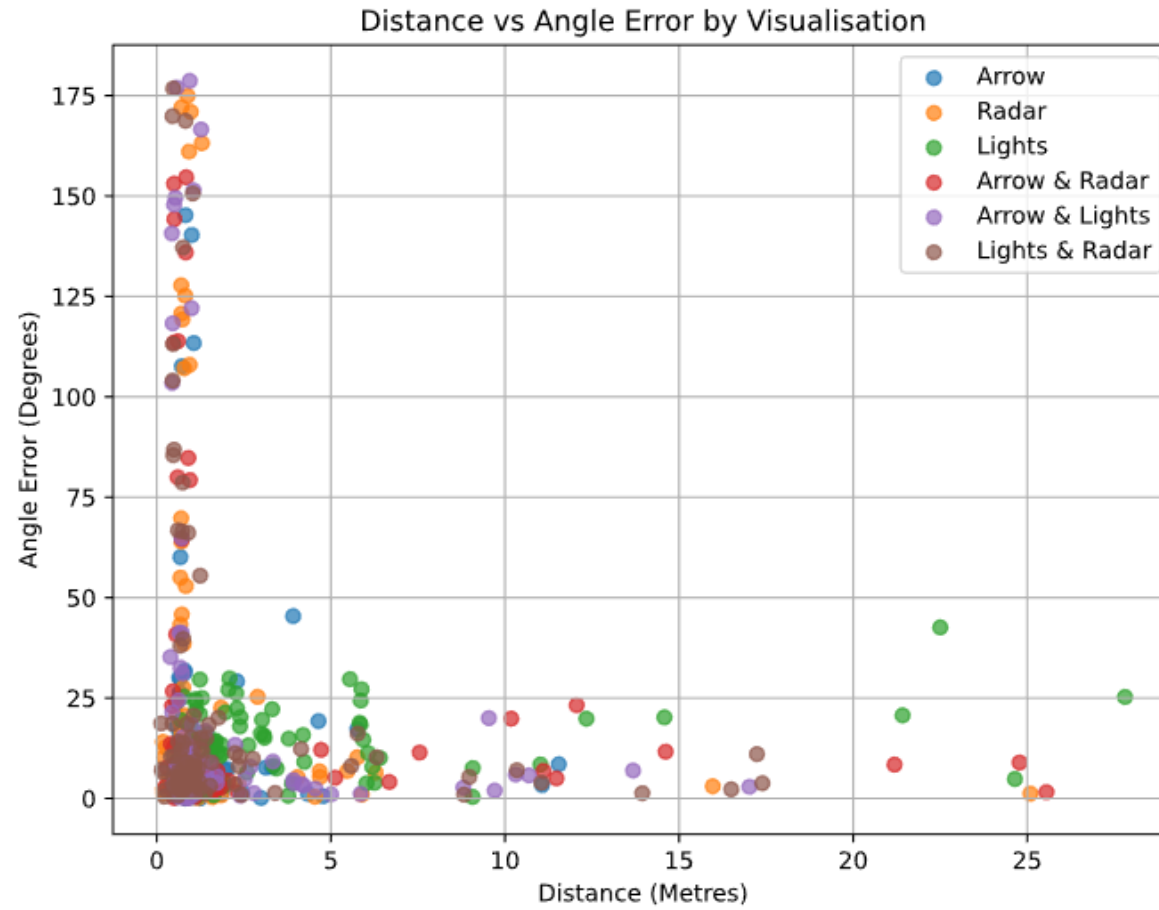
Angular Error Distribution by Visualisation



Distance to Target Distribution by Visualisation

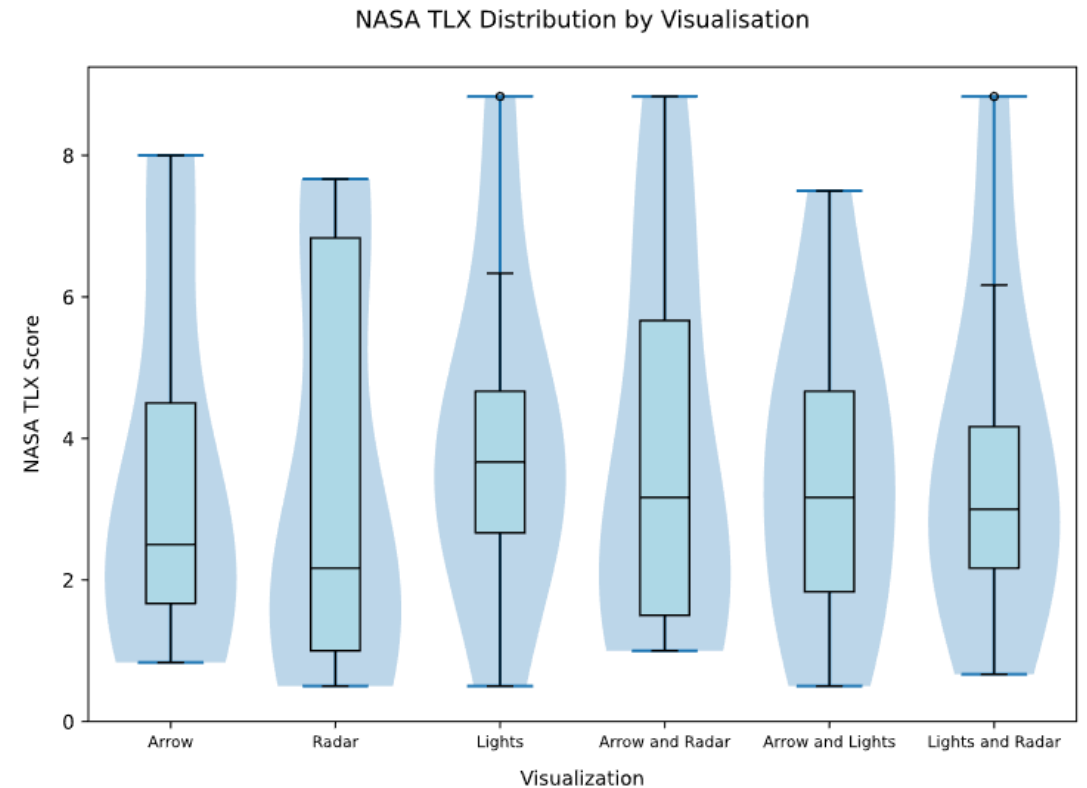


Plot of angle against Distance



Key Findings

- No single visualisation or combination works best
- Users preferred to be able to switch between visualisations depending on the task or distance from sources
- Usability similar across all visualisations
- Visual load higher for combinations
- Design implications:
 - Need to clearly encode arms reach distance
 - Users highlighted the need for sound classification to filter relevant sounds, such as warnings



Conclusion

- Spatial sound visualisation is context-dependent and subject to user preference
- More than 1 design is required, trade off between different visualisation techniques
- Systems should support adaptive, flexible visualisations
- A clear encoding of distance is important

Further Work

- Locating sound sources using the real-world prototype
- Visualising spatial sound in 3D instead of only the azimuth
- Visualising moving sounds
- Visualising multiple sounds at once

Any Questions?